Tzu-Chieh (Jeremy) Chao

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EDUCATION

University of Florida **Aug 2024 - May 2026**
Master of Science, Applied Data Science *Gainesville, FL*
- **Achievements:** Achievement Award Scholarship, Ranked in the top 1% of the department by academic performance.

National Taipei University **Sep 2017 - Jan 2022**
Bachelor of Business Administration, Business Administration *Taipei, Taiwan*
- **Coursework:** Linear Algebra, Probability and Statistics, Regression Analysis, Business Analytics

WORK EXPERIENCE

Aventusoft LLC **Aug 2025 - May 2026**
Machine Learning Engineer – UF IPPD Co-op *Gainesville & Boca Raton, FL*
- Collaborated with industry engineers via Git and Azure DevOps to define system architecture, software requirements, and hardware deployment constraints for a clinical-grade ECG AI platform.
- Formulated cloud and mobile inference deployment strategies using ONNX Runtime, CoreML, and TFLite to meet rigorous enterprise production specifications.
- Authored FDA Design Control documentation and verification protocols aligned with medical device software development practices (SaMD).
- Led cross-functional communication with executive sponsor liaisons, presenting technical deliverables, validation milestones, and dataset alignment across the project lifecycle.
- Trained and evaluated multi-task deep learning models for complex ECG analysis, optimizing classification and regression performance to achieve an F1-score of 93.5%, an AUC of 0.97, and a 15% reduction in MAE for fiducial metric extraction.

Commerce Development Research Institution **Sep 2022 - Jul 2023**
Business Analyst Consultant *Taipei, Taiwan*
- Analyzed sustainability and ESG disclosures from 30 Fortune 500 companies using pandas and scikit-learn to identify net-zero implementation patterns and business implications relevant to Taiwanese industries.
- Conducted data-driven operational impact analysis and developed data models to support competitiveness and strategy assessments for SMEs.
- Built ETL processes using SQL, dbt, and pandas to integrate sustainability data into a Snowflake analytical data warehouse on Azure, supporting exploratory analysis and Tableau dashboards for executive stakeholders.
- Partnered with senior research scientists to design analytical workflows and translate model-driven insights into staged, data-backed recommendations supporting Taiwan's 2050 net-zero emissions roadmap.

SELECTED PROJECTS

Electrocardiogram Deep Learning **Aug 2025 - May 2026**
University of Florida & Aventusoft, LLC
- Engineered robust preprocessing pipelines in Python (Pandas/NumPy) for 500 Hz single-lead ECG data, integrating bandpass filtering, sliding-window segmentation, and multi-source harmonization across LUDB, QTDB, PTB-XL, and CPSC datasets.
- Achieved a 1.5–6 ms fiducial timing error for P/QRS/T wave delineation by implementing a 1D U-Net segmentation model in PyTorch with Gaussian soft labels and morphology-aware post-processing.
- Evaluated lightweight neural networks for edge deployment, benchmarking a Transformer-based ECG Foundation Model (ECG-FM) against a custom CNN-BiLSTM classifier.
- Optimized on-device latency to <30 s and cloud inference to <2 min by executing model pruning and post-training quantization (PTQ) via TFLite and Docker containerization.

TECHNICAL SKILLS

- **Programming**: Python, C++, C, SQL, JavaScript, HTML, CSS, R, Excel VBA, Shell Scripting
- **Data Visualization**: Power BI, Tableau
- **Machine Learning & Data Science**: PyTorch, TensorFlow, Dbt, Snowflake, Databricks, Scikit-learn, Pandas
- **MLOps, Deployment & Cloud**: ONNX Runtime, CoreML, TFLite, FastAPI, Docker, Flask, MLOps, Git, CI/CD, APIs, Cloud Environments, Azure, Amazon S3